

Data Domains to Seed

1. Courses

To cover pricing and enrollment variations, I would create a representative course catalog including:

- Standard priced course (€100) for baseline validation
- Low-priced course (€5–10) for boundary testing
- High-priced course (€999+) for large transaction validation
- Free course to validate zero-value checkout
- Unpublished/draft course for negative scenarios
- Course restricted to specific coupons

Each course would have a unique identifier, defined price, and clear publication status.

2. Coupons

Since coupon logic is a medium to high-risk area, I would seed structured coupon configurations including:

- Percentage discounts (10%, 50%, 100%)
- Fixed discounts (€10, €500)
- Expired coupon
- Future-valid coupon
- Single-use coupon
- Limited-use coupon
- Course-specific coupon
- Cart-wide coupon
- Coupon exceeding cart value
- Non-stackable coupon (if supported)

Each coupon would include expiration rules, usage limits, scope, and discount caps.

3. Users

I would create user personas to validate different lifecycle states:

- New user (first purchase)
- Returning user
- User already enrolled in a course
- User with failed payment history
- Logged-out scenario validation

Each user would have a unique email and controlled purchase history.

4. Cart Configurations

Prepared cart states would include:

- Single course cart
 - Multi-course cart
 - Mixed pricing cart
 - Cart with coupon applied
 - Cart with expired coupon
 - Empty cart scenario
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5. Payment Sandbox States

Where supported, I would configure payment simulations for:

- Successful payment
- Declined payment
- Insufficient funds

- Timeout scenario
- Duplicate submission
- Payment retry flow

This ensures deterministic validation of payment handling.

Data Lifecycle Management

To maintain test environment stability:

- Use unique users per scenario where needed. If a scenario needs a fresh/new user we will create one, otherwise we will pick from an already existing pool of test users.
- Reset or isolate single-use coupons between runs
- Tag test orders for cleanup
- Periodically remove test artifacts. This could be done after each test execution or periodically with a cron job.
- Maintain versioned datasets